



Hubble 28th anniversary

Hubble Space Telescope celebrates 28th anniversary with a special treat to a Lagoon Nebula photo. <https://digit.in/hubble28>



New cancer tracking technology

Blood test using gold particles will give early prognosis of a cancer treatment's progress to oncologists. <https://digit.in/cancermon>

School of Medicine, determined that the interstitium is made up of liquid-filled collagen bundles. Further samples cemented the presence of this structure in tissues in multiple areas of the body.

This wasn't the first time in vivo microscopy had revealed a previously undiscovered anatomically significant part of the human body. In recent times, lymphatic vessels in the brain, were identified for the first time using in vivo multiphoton microscopy imaging through a thinned skull preparation. (<https://digit.in/LympBrn>)

WHAT DOES IT MEAN FOR THE HUMAN BODY?

Further examination from the researchers seems to indicate that there is a direct connection between the interstitial space and the lymphatic system. Apparently, the lymphatic fluid gets produced in the interstitium and flows into the lymphatic system. Hence, cancer tumors that get into this space can spread to lymph nodes. As the researchers have themselves put it, interstitium could be a veritable 'water



Dr Michael Nathanson

slide' for cancer. If this turns out to be true, it will shed new light on how cancer spreads in the human body.

This network was almost discovered in an older study. In 2011, Dr. Michael Nathanson and colleagues from Yale University School of Medicine

had published a paper highlighting their discovery of a network of dark fibers that they weren't able to figure out properly.

There are other purposes that the interstitium could be fulfilling as well, such as cushioning our organs, transporting white blood cells or molecules that are used to signal communication or more. It is the omnipresence of this network in our body that makes its potential capabilities so vast and diverse. That brings us to the important question – is it an organ?

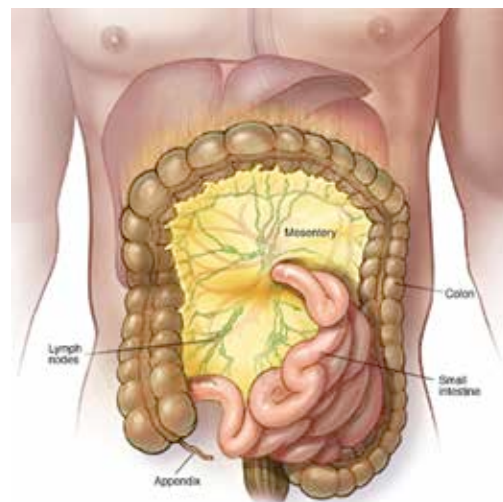
IS IT AN ORGAN?

If you remember, about a year ago, popular media went into a similar tizzy over mesentery, which was - you guessed it - a new organ. A fan-shaped bunch of tissues that holds our gut in place, we had known about mesentery for a long while but only recently discovered it to be large enough to warrant a discussion over its organ status. Many pieces reported it as the body's 79th organ. Truth be told, that number is entirely hogwash.

That is because there is no central consensus over the definition of an organ. Biologists over the world do not have one particular way of telling if something in your body is an organ or not. In fact, they still debate about the topic with no conclusion in sight. The most generic definition that is thrown up with a simple Google search is 'a part of an organism which is typically self-contained and has a specific vital function.' Even a somewhat cursory search beyond this provides you with enough room to stretch the definition as you see fit. Which is why parts of our body like skin, eyeballs, nipples and the likes of mesenteries have been classified and declassified as organs quite arbitrarily over the years.

Is it really necessary to have the definition for an organ? You bet it is. As we increasingly integrate technol-

ogy into healthcare and let artificial intelligence run havoc on medical data, we need to have clear and precise definitions for things. Algorithms running on medical data to identify newer ways of detecting diseases and ailments and finding novel ways to



Mesentery, the deja vu to Interstitium

cure the same need to know that an organ in one place is an organ in another. A disparity in this case would make such algorithms, and consequently our efforts to integrate AI into healthcare, increasingly problematic.

SO WHERE DOES THAT LEAVE US?

At the moment? Right where we started off. Our classification of the interstitium as an organ (or the lack of it thereof) will, in no way, affect its actual functioning and importance. It will continue to fulfil its currently somewhat elusive purpose regardless of the scientific community's ability to agree upon the definition of an organ.

Instead, what it does merit is further study into the literature that has acknowledged its existence in the past (there's plenty of it) without classifying it as an organ to understand its true nature. In essence, we need to find out what we already know about it without knowing that we do. Confused? Well, maybe it will turn out that your interstitium can help you deal with that. 